

Advanced (Habanero)

- Q1.** The vertex form of a quadratic function is $y = a(x-h)^2 + k$. What does h represent?
- (A) The y -intercept
(B) The x -coordinate of the vertex
(C) The axis of symmetry
(D) The maximum or minimum value
(E) The coefficient of x^2
- Q2.** Write the quadratic function $y = x^2 + 6x + 8$ in vertex form.
- (A) $y = (x + 3)^2 - 1$
(B) $y = (x + 4)^2 + 2$
(C) $y = (x + 2)^2 + 4$
(D) $y = (x + 1)^2 + 7$
(E) $y = (x - 3)^2 - 1$
- Q3.** What is the x -intercept of the quadratic function $y = (x - 2)(x + 5)$?
- (A) $(-5, 0)$
(B) $(-2, 0)$
(C) $(2, 0)$
(D) $(5, 0)$
(E) $(0, 0)$
- Q4.** The quadratic function $y = a(x - p)(x - q)$ is in intercept form. What do p and q represent?
- (A) The x -intercepts
(B) The y -intercepts
(C) The vertex coordinates
(D) The axis of symmetry
(E) The coefficient of x^2
- Q5.** Write the quadratic function $y = 2(x - 1)(x + 4)$ in vertex form.
- (A) $y = 2(x + 1.5)^2 - 10$
(B) $y = 2(x + 2.5)^2 - 15$
(C) $y = 2(x - 1.5)^2 + 10$
(D) $y = 2(x - 3)^2 - 8$
(E) $y = 2(x + 3)^2 - 16$
- Q6.** What is the axis of symmetry of the quadratic function $y = (x - 3)^2 + 2$?
- (A) $x = -3$
(B) $x = 0$
(C) $x = 3$
(D) $x = 2$
(E) $x = 1$
- Q7.** Write the quadratic function $y = x^2 - 4x - 3$ in intercept form.
- (A) $y = (x - 1)(x + 3)$
(B) $y = (x - 3)(x + 1)$
(C) $y = (x + 1)(x - 3)$
(D) $y = (x - 2)(x + 2)$
(E) $y = (x + 2)(x - 2)$
- Q8.** The quadratic function $y = (x - 2)^2 + 1$ has a vertex at (h, k) . What are the coordinates of the vertex?
- (A) $(2, 1)$
(B) $(1, 2)$
(C) $(0, 0)$
(D) $(1, 0)$
(E) $(2, 0)$

Open Ended Questions (FRQ)

- Q9.** Write the quadratic function $y = x^2 + 2x - 6$ in both vertex form and intercept form.
- Q10.** Find the x -intercepts and the vertex of the quadratic function $y = (x + 2)(x - 4)$. Give the coordinates of the vertex and the x -intercepts.

Chef's Tips

- Tip 1: Check for the vertex and intercept forms of the quadratic functions.
- Tip 2: Verify the coordinates of the vertex and the x -intercepts.
- Tip 3: Consider the axis of symmetry when graphing the quadratic functions.